

Indexing in Pandas

```
In [64]: import pandas as pd
```

```
In [65]: df = pd.read_csv(r"world_population.csv")

df
```

Out[65]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	Pop
0	36	AFG	Afghanistan	Kabul	Asia	41128771.0	38972230.0	33753499.0	28
1	138	ALB	Albania	Tirana	Europe	2842321.0	2866849.0	2882481.0	29
2	34	DZA	Algeria	Algiers	Africa	44903225.0	43451666.0	39543154.0	35
3	213	ASM	American Samoa	Pago Pago	Oceania	44273.0	46189.0	51368.0	
...	...	...	...	...	...	...	...	...	...
230	172	ESH	Western Sahara	El Aaiún	Africa	575986.0	556048.0	491824.0	4
231	46	YEM	Yemen	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24
232	63	ZMB	Zambia	Lusaka	Africa	20017675.0	18927715.0	NaN	13
233	74	ZWE	Zimbabwe	Harare	Africa	16320537.0	15669666.0	14154937.0	12

234 rows × 17 columns

```
In [66]: df = pd.read_csv(r"world_population.csv", index_col = "Country")

df
```

Out[66]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	2010 Population
Country								
Afghanistan	36	AFG	Kabul	Asia	41128771.0	38972230.0	33753499.0	2818967
Albania	138	ALB	Tirana	Europe	2842321.0	2866849.0	2882481.0	291335
Algeria	34	DZA	Algiers	Africa	44903225.0	43451666.0	39543154.0	3585634
American Samoa	213	ASM	Pago Pago	Oceania	44273.0	46189.0	51368.0	5484
...	...	...	...	...	...	...	...	...
Western Sahara	172	ESH	El Aaiún	Africa	575986.0	556048.0	491824.0	41325
Yemen	46	YEM	Sanaa	Asia	33696614.0	32284046.0	28516545.0	2474394
Zambia	63	ZMB	Lusaka	Africa	20017675.0	18927715.0	NaN	1379208
Zimbabwe	74	ZWE	Harare	Africa	16320537.0	15669666.0	14154937.0	1283977

234 rows × 16 columns



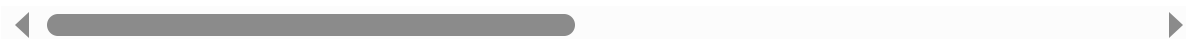
```
In [67]: df.reset_index(inplace=True) # inplace = True saves it
```

```
In [68]: df
```

Out[68]:

	Country	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	Po
0	Afghanistan	36	AFG	Kabul	Asia	41128771.0	38972230.0	33753499.0	28
1	Albania	138	ALB	Tirana	Europe	2842321.0	2866849.0	2882481.0	29
2	Algeria	34	DZA	Algiers	Africa	44903225.0	43451666.0	39543154.0	358
3	American Samoa	213	ASM	Pago Pago	Oceania	44273.0	46189.0	51368.0	
...	...	...	...	...	...	...	...	...	...
230	Western Sahara	172	ESH	El Aaiún	Africa	575986.0	556048.0	491824.0	4
231	Yemen	46	YEM	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24
232	Zambia	63	ZMB	Lusaka	Africa	20017675.0	18927715.0	NaN	13
233	Zimbabwe	74	ZWE	Harare	Africa	16320537.0	15669666.0	14154937.0	128

234 rows × 17 columns



```
In [69]: df.set_index('Country', inplace = True) # setting index column - country. inplace
df
```

Out[69]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	2010 Population
Country								
Afghanistan	36	AFG	Kabul	Asia	41128771.0	38972230.0	33753499.0	2818967
Albania	138	ALB	Tirana	Europe	2842321.0	2866849.0	2882481.0	291335
Algeria	34	DZA	Algiers	Africa	44903225.0	43451666.0	39543154.0	3585634
American Samoa	213	ASM	Pago Pago	Oceania	44273.0	46189.0	51368.0	5484
...	...	...	...	...	...	...	...	...
Western Sahara	172	ESH	El Aaiún	Africa	575986.0	556048.0	491824.0	41325
Yemen	46	YEM	Sanaa	Asia	33696614.0	32284046.0	28516545.0	2474394
Zambia	63	ZMB	Lusaka	Africa	20017675.0	18927715.0	NaN	1379208
Zimbabwe	74	ZWE	Harare	Africa	16320537.0	15669666.0	14154937.0	1283977

234 rows × 16 columns



In [70]: `df.loc['Albania']`

Out[70]: Rank 138
CCA3 ALB
Capital Tirana
Continent Europe
...
Area (km²) 28748.0
Density (per km²) 98.8702
Growth Rate 0.9957
World Population Percentage 0.04
Name: Albania, Length: 16, dtype: object

In [71]: `df.iloc[1]`

Out[71]: Rank 138
CCA3 ALB
Capital Tirana
Continent Europe
...
Area (km²) 28748.0
Density (per km²) 98.8702
Growth Rate 0.9957
World Population Percentage 0.04
Name: Albania, Length: 16, dtype: object

In [72]: `df.reset_index(inplace = True)` *# reset index , inplace = True saves it*

```
In [73]: df
```

Out[73]:

	Country	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	Po
0	Afghanistan	36	AFG	Kabul	Asia	41128771.0	38972230.0	33753499.0	28
1	Albania	138	ALB	Tirana	Europe	2842321.0	2866849.0	2882481.0	29
2	Algeria	34	DZA	Algiers	Africa	44903225.0	43451666.0	39543154.0	358
3	American Samoa	213	ASM	Pago Pago	Oceania	44273.0	46189.0	51368.0	
...	...	...	...	...	...	...	...	...	
230	Western Sahara	172	ESH	El Aaiún	Africa	575986.0	556048.0	491824.0	4
231	Yemen	46	YEM	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24
232	Zambia	63	ZMB	Lusaka	Africa	20017675.0	18927715.0	NaN	13
233	Zimbabwe	74	ZWE	Harare	Africa	16320537.0	15669666.0	14154937.0	128

234 rows × 17 columns

```
In [74]: df.set_index(['Continent','Country'], inplace=True) # setting index
```

```
In [75]: df
```

Out[75]:

		Rank	CCA3	Capital	2022 Population	2020 Population	2015 Population	2010 Population
Continent	Country							
Asia	Afghanistan	36	AFG	Kabul	41128771.0	38972230.0	33753499.0	2818967
Europe	Albania	138	ALB	Tirana	2842321.0	2866849.0	2882481.0	291335
Africa	Algeria	34	DZA	Algiers	44903225.0	43451666.0	39543154.0	3585634
Oceania	American Samoa	213	ASM	Pago Pago	44273.0	46189.0	51368.0	5484
...	...	...	...	...	...	...	...	...
Africa	Western Sahara	172	ESH	El Aaiún	575986.0	556048.0	491824.0	41325
Asia	Yemen	46	YEM	Sanaa	33696614.0	32284046.0	28516545.0	2474394
Africa	Zambia	63	ZMB	Lusaka	20017675.0	18927715.0	NaN	1379208
	Zimbabwe	74	ZWE	Harare	16320537.0	15669666.0	14154937.0	1283977

234 rows × 15 columns



In [76]: `df.sort_index()` *# sort_index() - grouping continents (index)*

Out[76]:

		Rank	CCA3	Capital	2022 Population	2020 Population	2015 Population	2010 Population
Continent	Country							
Africa	Algeria	34	DZA	Algiers	44903225.0	43451666.0	39543154.0	3585634
	Angola	42	AGO	Luanda	35588987.0	33428485.0	28127721.0	23364
	Benin	77	BEN	Porto-Novo	13352864.0	12643123.0	10932783.0	9445
	Botswana	144	BWA	Gaborone	2630296.0	2546402.0	2305171.0	2091
...	...	...	...	...	...	...	...	...
South America	Peru	44	PER	Lima	34049588.0	33304756.0	30711863.0	29225
	Suriname	170	SUR	Paramaribo	618040.0	607065.0	575475.0	546
	Uruguay	133	URY	Montevideo	3422794.0	3429086.0	3402818.0	3352
	Venezuela	51	VEN	Caracas	28301696.0	28490453.0	30529716.0	28715

234 rows × 15 columns



```
In [77]: pd.set_option('display.max.rows', 8)      #if we want to see 8 rows
```

```
In [78]: df
```

Out[78]:

		Rank	CCA3	Capital	2022 Population	2020 Population	2015 Population	2010 Population
Continent	Country							
Asia	Afghanistan	36	AFG	Kabul	41128771.0	38972230.0	33753499.0	28189670.0
Europe	Albania	138	ALB	Tirana	2842321.0	2866849.0	2882481.0	2913390.0
Africa	Algeria	34	DZA	Algiers	44903225.0	43451666.0	39543154.0	35856340.0
Oceania	American Samoa	213	ASM	Pago Pago	44273.0	46189.0	51368.0	54840.0
...	...	...	...	...	...	...	...	...
Africa	Western Sahara	172	ESH	El Aaiún	575986.0	556048.0	491824.0	413290.0
Asia	Yemen	46	YEM	Sanaa	33696614.0	32284046.0	28516545.0	24743940.0
Africa	Zambia	63	ZMB	Lusaka	20017675.0	18927715.0	NaN	13792000.0
	Zimbabwe	74	ZWE	Harare	16320537.0	15669666.0	14154937.0	12839700.0

234 rows × 9 columns

```
In [79]: df.loc['Africa', 'Angola']
```

Out[79]:

Rank	42
CCA3	AGO
Capital	Luanda
2022 Population	35588987.0
...	...
Area (km²)	1246700.0
Density (per km²)	28.5466
Growth Rate	1.0315
World Population Percentage	0.45
Name: (Africa, Angola), Length: 15, dtype: object	

```
In [80]: df.iloc[1]
```

```
Out[80]: Rank                138
          CCA3                ALB
          Capital            Tirana
          2022 Population    2842321.0
          ...
          Area (km²)        28748.0
          Density (per km²)  98.8702
          Growth Rate       0.9957
          World Population Percentage  0.04
          Name: (Europe, Albania), Length: 15, dtype: object
```

```
In [81]: df.loc['Africa']
```

Out[81]:

	Rank	CCA3	Capital	2022 Population	2020 Population	2015 Population	2010 Population	Popul
Country								
Algeria	34	DZA	Algiers	44903225.0	43451666.0	39543154.0	35856344.0	30774
Angola	42	AGO	Luanda	35588987.0	33428485.0	28127721.0	23364185.0	16394
Benin	77	BEN	Porto- Novo	13352864.0	12643123.0	10932783.0	9445710.0	6998
Botswana	144	BWA	Gaborone	2630296.0	2546402.0	2305171.0	2091664.0	1726
...	...	...	...	...	...	...	...	...
Uganda	31	UGA	Kampala	47249585.0	44404611.0	37477356.0	32341728.0	24020
Western Sahara	172	ESH	El Aaiún	575986.0	556048.0	491824.0	413296.0	270
Zambia	63	ZMB	Lusaka	20017675.0	18927715.0	NaN	13792086.0	9891
Zimbabwe	74	ZWE	Harare	16320537.0	15669666.0	14154937.0	12839771.0	11834

57 rows × 15 columns